CUSUM RAP Tool: User Manual

1. Introduction

Welcome to the CUSUM RAP (Reproducible Analytical Pipeline) Tool.
This application is designed to help public health analysts and epidemiologists detect early-stage outbreaks. By applying a CUSUM (Cumulative Sum) algorithm to your weekly aggregated case counts, the tool identifies areas where reported cases exceed the expected baseline, flagging potential anomalies for immediate review.

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2. Getting Started: Loading Your Data

Before running the analysis, you must input your epidemiological data. The tool accepts CSV files previously aggregated by the API-POP pipeline.

1. Navigate to the top menu bar and click **Files**.
2. Select **Upload RAW Data** to upload a new CSV file from your computer.
    - *Note on Overlaps:* If you upload multiple files that contain data for the same locations and weeks, the system will ask you how to handle the overlap. You can choose to keep the newest data, the oldest data, or sum the values together.

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3. Step-by-Step Configuration

Once your data is loaded, use the left-hand sidebar to configure the parameters for your analysis.

A. Geographic Selection

- **Geographic Level:** Select the administrative level you want to analyze (e.g., Country, Province, Department). The system will aggregate your weekly counts to match this level.
- **Specific Locations:** Click the dropdown to filter the analysis to specific regions. You can select multiple locations or leave it blank to evaluate all available areas.

B. Baseline Configuration

The CUSUM algorithm compares recent data against a historical baseline to detect anomalies.

- **Detection Period (Weeks):** Define how many recent weeks you want to actively monitor for outbreaks (minimum 52 weeks). The data *before* this period acts as your historical baseline.
- **Expected Frequency (μ_0):** This represents the normal, expected case count.
  - **Automatic (Poisson GLM):** The tool automatically calculates the expected baseline from your historical data.
  - **Manual Input:** Choose this if you want to set a strict, predefined reference frequency (measured per 100,000 people).

C. CUSUM Parameters

These statistical parameters dictate how sensitive the alarm system is.

- **Target ARL0 (Average Run Length):** The expected number of weeks before a false alarm occurs. A higher number means fewer false alarms but may delay the detection of a real outbreak.
- **RR (Relative Risk):** The magnitude of an outbreak you want to detect (e.g., a 20% increase in cases would be an RR of 1.20). *The tool will automatically calculate the optimal reference value (k) based on this input.*
- **h (Threshold):** The decision limit. If the accumulated deviations exceed this number, an alarm is triggered. *Once you input your ARL0 and RR, the tool will display a Recommended h value just below this box.*

D. Run Analysis

Once all parameters are set, click the **Run Analysis** button at the bottom of the sidebar.

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4. Reviewing the Results

After the analysis completes, use the main panel tabs to explore the findings.

Overview Tab (Heatmap)

This tab provides a high-level visual summary of all selected locations. The heatmap highlights specific weeks where the CUSUM threshold was breached, allowing you to quickly spot regional clusters or widespread anomalies.

Detailed View Tab

Dive into specific areas using the **Select Location** dropdown.

- **Bar Plot:** Displays the raw observed case counts versus the expected baseline.
- **Trends Plot:** Visualizes the CUSUM process itself, showing how the cumulative sum approaches and breaks the threshold limit (h).
- *Export:* Use the yellow and blue download buttons to save these charts as PNG images for your reports.

Analysis Results Tab

View the raw mathematical output in a comprehensive data table. This table includes the exact statistics, k values, and alarm triggers for every location and week. Click **Download Results CSV** to export this data for further use in Excel or other analytical software.

Prepared Data Tab

This tab shows a **summary of the cleaned dataset** used for the analysis.
It includes:

- Number of geographic units at each level (based on the configuration)
- Total number of weeks in the time series

Note: The full dataset is not displayed; this tab is for quick validation only.